

Veterinary Package Insert

TP-FLUBEN 50MG/G POWDER (5% Flubendazole)

PRODUCT DESCRIPTION

White or almost white colour powder, which contains 50mg of flubendazole per gram in 500gm or 1kg sachet.

PHARMACODYNAMICS & PHARMACOKINETICS

Flubendazole acts by binding to tubulin, the dimeric subunit protein of the microtubules. It inhibits microtubular assembly in absorptive cells: i.e. of intestinal cells of nematodes or the tegumental cells of cestodes. This is shown by disappearance of cytoplasmic microtubules, accumulation of secretory granules in the cytoplasm due to a block in their transport, leading to an impaired coating of the cellular membrane and a decreased digestion and absorption of nutrients. Irreversible lytic degeneration of the cell, due to the accumulation of secretory substances (hydrolytic and proteolytic enzymes), results in the death of the parasite.

These changes are relatively fast and are primarily seen in those organelles directly involved in the secretory and absorptive functions of the cells. In contrast the changes are not seen in host cells.

Flubendazole is very poorly soluble in aqueous systems, such as the gastrointestinal tract, which results in a low dissolution rate and a very low absorption. This is reflected by a high faecal excretion of unchanged parent drug. The very small fraction absorbed is extensively metabolised by first-pass metabolism in the liver, involving carbamate hydrolysis and ketone reduction. The biotransformation products are conjugated to glucuronides or sulphate conjugates and excreted in the bile and the urine.

The excretion in urine is relatively low and consists almost exclusively of metabolites with only small amounts of unchanged compound. In pigs, highest tissue levels are measured in liver and kidneys. The half-life of flubendazole in tissues is 1 to 2 days.

INDICATION

Flubendazole is a broad spectrum anthelmintic, effective against mature and immature stages of the following nematodes of the gastro-intestinal tract:

Ascaris suum, (large roundworm) including migrating larvae, *Hyostrongylus rubidus*, (red

stomach worm), *Oesophagostomum dentatum* (nodular worm), *Trichuris suis* (whipworm) and *Strongyloides ransomi* (threadworm) (adult).

Flubendazole is ovicidal.

TARGET SPECIES

Pigs

RECOMMENDED DOSAGE AND ADMINISTRATION

To ensure administration of a correct dose, body weight should be determined as accurately as possible.

If animals are to be treated collectively rather than individually, they should be grouped according to their bodyweight and dosed accordingly, in order to avoid under- or overdosing.

Individual treatment (single administration):

i. Dosage:

Add 1 g of the veterinary medicinal product for each 10 kg bodyweight onto the finished feed, as a single animal dose. (This is equivalent to 5 mg of flubendazole per one kg bodyweight).

ii. Treatment frequency:

Twice a year unless recommended otherwise by a veterinary surgeon. Pigs brought onto the premises should be treated on arrival and before mixing with other animals. Regular faecal examination is advocated to know which worms are present on the farm so that specific measures may be taken to prevent re-infection.

iii. Treatment of clinical worm infestations:

Treat relevant infestations at the following intervals:

Nodular worm (*Oesophagostomum dentatum*)

- every 2 months

Large roundworm (*Ascaris suum*)

- every 2 months

Red stomach worm (*Hyostrongylus rubidus*)

- every month

Whipworm (*Trichuris suis*)

- every 6 weeks.

CONTRAINDICATIONS

Do not use in case of hypersensitivity to the active substance or to any of the excipients.

WARNINGS & PRECAUTIONS

Special warnings for each target species:

Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.

Underdosing, which may be due to underestimation of body weight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

Accidental ingestion by humans should be avoided. May cause sensitisation by skin contact. May cause skin and eye irritation. Wear overalls, safety glasses and impervious gloves when mixing and handling the product. Wash affected parts if skin contact occurs. If accidental eye contact occurs, immediately rinse thoroughly with water. If the operation involves potential exposure to dust, wear either a disposable filter on a half mask respirator.

INTERACTIONS WITH OTHER MEDICATIONS

None known.

PREGNANCY AND LACTATION

The product may be used in pregnant and lactating animals.

ADVERSE EFFECTS

None known.

OVERDOSE & TREATMENT

Flubendazole has a low acute oral toxicity and is well tolerated in the target species. In situations where accidental overdose is suspected of having occurred, there is no antidote and treatment should be symptomatic.

WITHDRAWAL PERIOD

Pigs must not be slaughtered for human consumption during treatment.

Pigs may be slaughtered for human consumption only after 7 days from the last treatment.

MAXIMUM RESIDUAL LIMIT (MRL)

Substance	Food	Maximum Residue Limits (MRLs) in food µg/kg
Flubendazole	Muscle, liver (pig)	10

	Fat (pig)	20
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STORAGE CONDITION

Store below 30°C. Protect from direct sunlight.

SHELF LIFE

2 years.

After 1st opening, reconstitution or dilution, use within 24 hours.

PACKING

600gm

1kg

15kg

THYE PHARMA SDN. BHD.

(manufacturer & product registration holder)
No.2 Lorong Industri Ringan Permatang
Tinggi 8,
Kawasan Industri Ringan Permatang Tinggi,
14000 Bukit Mertajam,
Pulau Pinang.

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